

1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

	DDR3	DDR4	DDR5
Clock Rate	400–1066 MHz	800–1600 MHz	2400–3600 MHz
Voltage	1.5 V	1.2 V	1.1 V
Max Size	8GB	32GB	128GB
Channel Architecture	Single 64-bit channel	Single 64-bit channel	Two independent 32-bit channels
Typical Use Cases	Legacy systems, office PCs, light web browsing	Mainstream desktops, budget gaming, general multitasking	High-end gaming, content creation, AI/ML tasks, servers
DIMM Pins	240 (R, LR, U)	288 (R, LR, U)	288 (R, LR, U)

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint- Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

DONE

<https://youtu.be/kRMJwiXhrEU?si=1WhHkCYyi-X-Hqcb>

3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

	Hard Disk Drive (HDD)	Solid State Drive (SSD)	Non-Volatile Memory Express (NVMe)
Technology	Mechanical. Uses spinning magnetic disks (platters) and an actuator arm to read/write data.	Contains no moving parts, resulting in higher durability and silence.	Advanced Flash memory connecting directly to the motherboard via the PCIe bus , allowing for simultaneous parallel processing of millions of commands.
Speed	Slowest of the three	Faster, typically utilizing the SATA interface	ranging from 3,000 to 7,000+ MB/s
Cost	Lowest in price , making it ideal for massive storage.	more expensive than HDDs but significantly cheaper than NVMe.	Most expensive storage type per gigabyte
Daily App	Google Drive or Apple Time Machine to archive large files or movies perfectly.	Microsoft Windows or Google Chrome installation on an SSD greatly reduces boot times and app response compared to an HDD	Adobe Premiere Pro or Steam require rapid loading and exporting of massive high-resolution files

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

DONE

<https://youtu.be/i2zLpN-bfk0?si=8-ezf0RuPKtpjSd>